

HEAVY VEHICLE FAILURE PREDICTION THROUGH INTELLIGENT SENSOR DATA

Project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING BACHELOR OF TECHNOLOGY

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Abstract

Heavy vehicles operate under demanding conditions, and unexpected component failures can cause downtime, maintenance delays, increased operating costs, and safety risks. This project develops a machine-learning approach for predicting Air Pressure System related failures in Scania heavy trucks using sensor and operational data. The dataset contains 76,000 records and 170 anonymized sensor features, together with a binary target indicating whether a failure is related to the APS component. The data contains substantial missing values and a highly imbalanced target distribution. The project therefore applies missing-value handling, stratified splitting, leakage-safe pipelines, class-weighted classification, exploratory data analysis, hyperparameter tuning, and model comparison. Logistic Regression, Decision Tree, and Random Forest models are evaluated using accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC, and confusion matrices. Recall is emphasized because failing to detect an actual vehicle failure can be more costly than an unnecessary inspection.

Keywords: predictive maintenance; heavy vehicles; APS failure; sensor data; imbalanced classification; Random Forest

1. Introduction

1.1 Purpose of the Project

Heavy-duty trucks are essential to transportation and logistics. Failure of a critical component can produce unplanned downtime, maintenance expenses, and safety concerns. The Air Pressure System is particularly important because pressurized air supports functions such as braking and gear changes. Sensor and operational records collected during vehicle use provide an opportunity to identify patterns associated with failures before they become serious.

This project applies machine learning to support proactive inspection and maintenance prioritization. A reliable warning system can help maintenance teams focus attention on vehicles whose sensor patterns are associated with APS-related failure, while reducing dependence on fixed service schedules alone.

1.2 Problem Statement

The problem is a binary classification task. Given anonymized sensor and operational measurements from a heavy vehicle, the system predicts whether the observed failure is related to the Air Pressure System. The expected output is a binary class: 0 for a non-APS-related failure and 1 for an APS-related failure.

1.3 Existing Methods and Disadvantages

Traditional maintenance often relies on fixed schedules or reacts after a component has failed. These approaches may inspect healthy vehicles unnecessarily and may fail to identify a developing problem early. Rule-based systems can also become difficult to maintain when many sensor variables interact.

A further difficulty is the strong class imbalance in the supplied data. A model that optimizes accuracy alone may mostly predict the majority class and still miss actual APS failures. Therefore, the project uses class-weighted models and evaluation metrics that measure minority-class detection.

1.4 Proposed System

The proposed system loads the APS Scania dataset, converts the na representation into missing values, checks duplicates, performs exploratory analysis, splits the data using stratification, and uses leakage-safe preprocessing pipelines. Logistic Regression, Decision Tree, and Random Forest models are trained and compared.

RandomizedSearchCV is then used to tune the Random Forest with recall as the primary search metric. The final model produces a class prediction and failure probability for unseen sensor records.

1.5 Research Objectives

1. Prepare and describe the APS Scania sensor dataset.
2. Perform systematic preprocessing and exploratory analysis.
3. Retain relevant sensor features while preventing data leakage.
4. Develop and train suitable machine-learning classification models.
5. Evaluate and compare models using appropriate imbalanced-classification metrics.
6. Identify a final model and discuss its practical applicability.

2. Literature Survey

Recent studies show that predictive maintenance research increasingly combines sensor preprocessing, imbalance handling, cost-sensitive learning, ensemble models, and explainability. The studies below were selected for their relevance to vehicle diagnostics, Scania data, sensor-based failure prediction, or imbalanced predictive maintenance.

S.No.	Title	Reference with DOI	Problem or gap	Objective	Methods	Inference	Future research
1	Cost-Optimised Machine Learning Model Comparison for Predictive Maintenance	Yang and Iqbal, Electronics, 2025, 14, 2497. DOI: 10.3390/electronics14122497	Imbalanced Scania Component X data and costly misclassification	Compare predictive-maintenance algorithms under cost-sensitive evaluation	SVM, RF, DT, KNN, MLP, XGBoost, LightGBM; sliding windows; downsampling and class weighting	Random Forest and SVM were strong; voting improved cost efficiency	Explore end-to-end failure prediction and cost-aware ensembles
2	Cost-Optimised Machine Learning Model Comparison for Predictive Maintenance	Dimidov, Jafarnejad and Frank, IEEE ICPHM, 2025; DOI not listed in supplied review	Simplified temporal representation may lose information	Reduce predictive-maintenance costs with compact features	Recent observations, transformed time-series data and AutoML	AutoML reduced maintenance costs while simplifying modelling	Use richer temporal and deep-learning representations
3	Enhancing Predictive Maintenance for Heavy Vehicles: A Multi-Stage Feature Selection and Preprocessing Framework on the Scania Dataset	Ferrisi, Conte, Guido and Ambrogio, Procedia Computer Science, 2026, 2514-2523. DOI: 10.1016/j.procs.2026.02.288	Missing values, anonymized variables, imbalance and feature redundancy	Improve robustness through multi-stage preprocessing and selection	Multi-stage preprocessing, feature selection and multiple ML models	Preprocessing and selection can improve transparency and predictive performance	Evaluate more models and generalization settings
4	Two-Stage LightGBM Framework for Cost-Sensitive Prediction of Impending Failures of Component X in Scania Trucks	Kim and Kim, Computers, Materials & Continua, 2026, 86(3), 50. DOI: 10.32604/cmc.2025.073492	Noisy, irregular and imbalanced Scania time series	Prioritize safety-critical failure detection and cost efficiency	Last-k statistical summaries and two-stage cost-sensitive LightGBM	High recall for safety-critical faults with lower misclassification cost	Improve intermediate-fault detection and adaptive thresholds
5	SCANIA Component X Dataset: A Real-World Multivariate Time Series Dataset for Predictive Maintenance	Kharazian, Lindgren, Magnússon, Steinert and Andersson Reyna, Scientific Data, 2025, 12, 493. DOI: 10.1038/s41597-025-04802-6	Need for realistic, multivariate predictive-maintenance benchmarks	Document a real-world heavy-vehicle time-series dataset	Dataset construction, vehicle records and multivariate sensor signals	Provides a realistic benchmark for imbalanced failure prediction	Study vehicle-level validation and temporal drift
6	Enhancing Predictive Maintenance in Automotive Industry: Addressing Class Imbalance Using Advanced Machine Learning Techniques	Mahale, Kolhar and More, Discover Applied Sciences, 2025, 7, 340. DOI: 10.1007/s42452-025-06827-3	Rare failure events and imbalance in automotive data	Improve minority-class detection	Resampling, class-imbalance analysis and ML/DL comparisons	Precision and recall are more informative than accuracy alone	Evaluate robust imbalance strategies on external fleets
7	Predictive Maintenance in Industry 4.0: A Survey of Planning Models and Machine Learning Techniques	Hector and Panjanathan, PeerJ Computer Science, 2024, 10, e2016. DOI: 10.7717/peerj-cs.2016	Many possible ML methods and maintenance objectives	Summarize planning and algorithm choices for PdM	Survey of supervised, unsupervised and semi-supervised methods	Algorithm choice depends on data type, objective and deployment context	Connect model selection with maintenance decisions
8	Machine Learning Algorithms for Predictive Maintenance in	Shah, International Journal of Engineering and Computer Science,	Vehicle health monitoring requires sensor-based prediction	Review ML algorithms for vehicle maintenance	Comparative discussion of ML and sensor-based diagnosis	Vehicle diagnostics benefits from combining sensor processing with ML	Validate on real fleets and investigate explainability

S.No.	Title	Reference with DOI	Problem or gap	Objective	Methods	Inference	Future research
	Autonomous Vehicles	2024, 13(01), 26015-26032. DOI: 10.18535/ijecs/v13i01.4786					
9	A Conceptual Framework for Machine Learning Algorithm Selection for Predictive Maintenance	Arena, Florian, Sgarbossa, Sølvsberg and Zennaro, Engineering Applications of Artificial Intelligence, 2024, 133, 108340. DOI: 10.1016/j.engappai.2024.108340	No single ML algorithm is best for every PdM problem	Guide practitioners in algorithm selection	Framework based on dataset, learning objective, accuracy and interpretability	Model selection should reflect data and maintenance objectives	Apply decision frameworks to more industrial datasets
10	An Integrated Deep Learning Approach for Predictive Vehicle Maintenance	Decision Analytics Journal, 2025, 16, 100597. DOI: 10.1016/j.dajour.2025.100597	Vehicle OBD data are temporal and partly unlabeled	Use temporal learning for vehicle maintenance	LSTM combined with clustering for OBD time series	Temporal representation can reveal hidden vehicle-health patterns	Compare temporal models with cost-sensitive tabular ensembles

3. Methodology

3.1 Proposed System Flow Diagram

Proposed System Flow for APS Failure Prediction

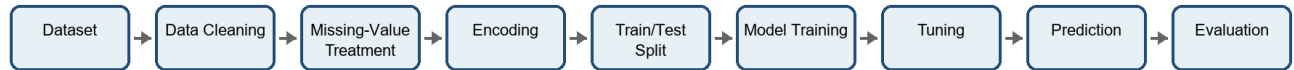


Figure 1. Proposed system flow for APS-related failure prediction.

3.2 Dataset Description

The project uses the APS Failure at Scania Trucks dataset supplied with the project files. The combined dataset contains 76,000 records and 171 columns: 170 anonymized sensor features and the target variable class. The target contains 74,625 negative records (98.19%) and 1,375 positive records (1.81%).

Property	Value
Records	76,000
Total columns	171
Sensor features	170
Target	class
Negative class	74,625 (98.19%)
Positive class	1,375 (1.81%)
Duplicate rows	0
Columns with missing values	169
Missing cells	1,078,695

3.3 Experimental Setup

Component	Configuration
Environment	Jupyter Notebook in Visual Studio Code
Operating system	Windows
Python	3.14.6
Libraries	Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Joblib
Split	80% training and 20% testing, stratified
Random state	42

3.4 Data Preprocessing

- Data inspection and schema validation were performed before modelling.
- The na representation was converted into missing values.
- No duplicate rows were found.
- Numerical missing values were treated with median imputation learned from training data.
- Outliers were retained because extreme values may represent genuine abnormal operating conditions.
- The target was encoded as 0 for neg and 1 for pos.
- StandardScaler was used for Logistic Regression; tree models do not require scaling.
- Pipelines were used to prevent data leakage.

3.5 Exploratory Data Analysis

EDA included descriptive statistics, target-class distribution, missingness analysis, representative sensor distributions, outlier inspection, and correlation analysis. The positive class is highly under-represented, and several sensor features have substantial missingness. These observations justify class-weighted learning and training-only imputation.

Target Class Distribution

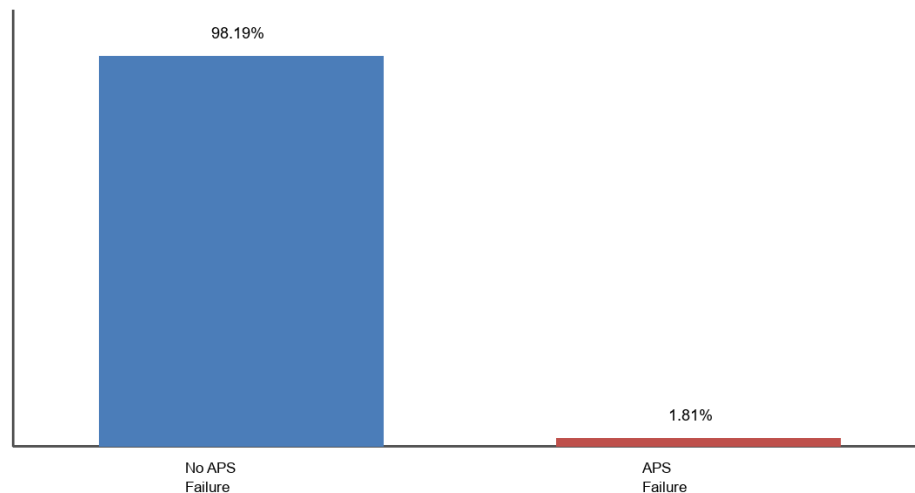


Figure 2. Imbalanced target-class distribution.

Top Sensor Features by Missing Percentage

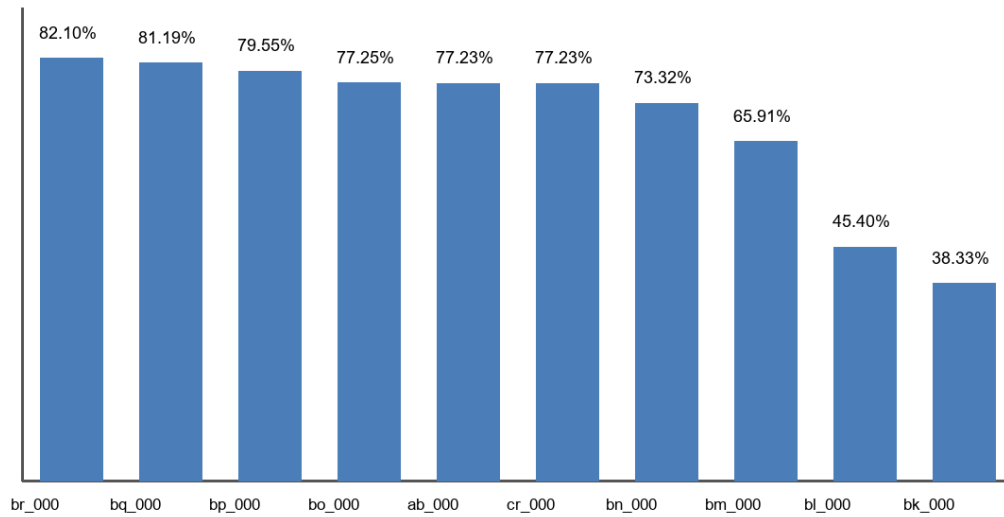


Figure 3. Top ten sensor features by missing percentage.

3.6 Feature Engineering and Feature Selection

No artificial features were created because the sensor names are anonymized and their physical meanings are not available. All 170 sensor features were retained after cleaning. This conservative choice avoids removing a low-variance sensor that may still provide useful information about an abnormal vehicle condition.

3.7 Model Building and Training

Logistic Regression was used as a linear baseline. Decision Tree was used as an interpretable nonlinear model. Random Forest was used as an ensemble model that combines many trees and can capture complex relationships among high-dimensional sensor variables. Class-weighted training was used to give greater importance to the minority failure class.

Model	Purpose	Configuration
Logistic Regression	Linear baseline	Balanced class weight, median imputation, standard scaling
Decision Tree	Interpretable nonlinear model	Balanced class weight, max depth 12, minimum leaf size 5
Random Forest	Ensemble nonlinear model	200 trees, balanced subsampling, parallel training

3.8 Hyperparameter Tuning

RandomizedSearchCV with three-fold stratified cross-validation was applied to the Random Forest pipeline. The search considered the number of trees, maximum depth, minimum leaf size, and maximum features. Recall was selected as the primary search metric because the project prioritizes detection of actual APS failures.

Tuning item	Value
Search method	RandomizedSearchCV
Cross-validation	3-fold stratified
Primary metric	Recall
Search parameters	n_estimators, max_depth, min_samples_leaf, max_features

3.9 Model Evaluation

The test set was held out from model fitting. Accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC, classification reports, and confusion matrices were used. Recall was emphasized because it measures the proportion of actual APS failures detected by the model.

4. Results and Discussion

4.1 Preprocessing Results

Stage	Result
Initial dataset	76,000 rows and 171 columns
Duplicate check	0 duplicate rows
Training set	60,800 rows and 170 features
Testing set	15,200 rows and 170 features
Missing-value treatment	Median imputation learned on training data
Feature count	170 sensor features retained

4.2 Model Performance

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	PR-AUC
Random Forest	99.1842%	93.1429%	59.2727%	72.4444%	99.0986%	88.3909%
Tuned Random Forest	97.4737%	41.1382%	92.0000%	56.8539%	99.0012%	76.0587%
Logistic Regression	97.4079%	40.4800%	92.0000%	56.2222%	97.5091%	78.2203%
Decision Tree	97.2368%	38.4737%	88.0000%	53.5398%	88.7283%	47.2331%

Random Forest achieved the best overall accuracy, precision, F1-score, ROC-AUC, and PR-AUC. Tuned Random Forest and Logistic Regression achieved the highest recall at 92.00%. The Tuned Random Forest was selected as the final model because the project gives priority to detecting actual APS failures.

4.3 Visual Results and Discussion

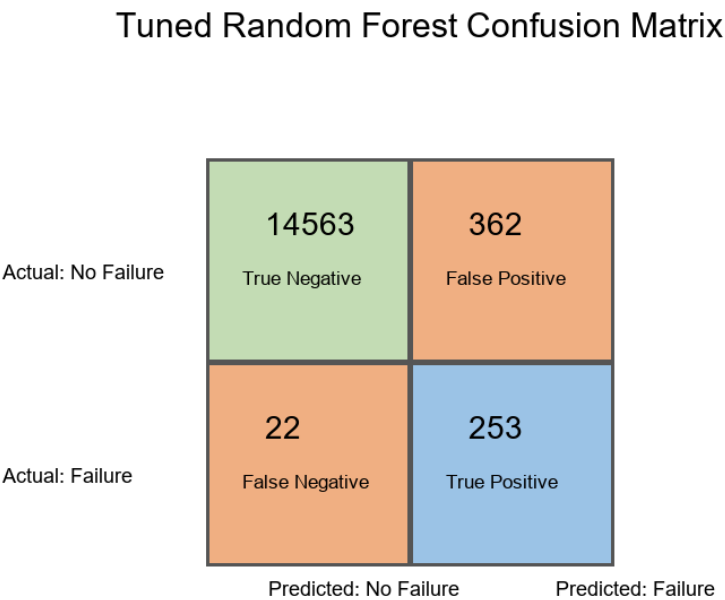


Figure 4. Confusion matrix for the Tuned Random Forest.

For the Tuned Random Forest, 253 actual failures were detected and 22 actual failures were missed. The model also produced 362 false positives, meaning that some healthy vehicles were flagged for inspection. In this application, false positives are more acceptable than false negatives because missing a real APS failure can create safety and maintenance risks.

The results demonstrate a trade-off. The untuned Random Forest provides higher precision and F1-score, while the Tuned Random Forest provides higher recall. The final selection therefore reflects the practical cost of missed failures rather than accuracy alone.

5. Conclusion

This project developed a machine-learning workflow for predicting APS-related failures in heavy Scania vehicles. The workflow combined data inspection, missing-value analysis, stratified splitting, leakage-safe preprocessing, class-weighted model training, hyperparameter tuning, and evaluation using metrics appropriate for imbalanced classification. The Tuned Random Forest achieved 92.00% recall and was selected because it detected most actual APS failures. The system can support inspection prioritization, but it should not replace professional maintenance decisions.

7. References

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